

# GULSHAN YADAV

AI ENGINEER | PYTHON BACKEND DEVELOPER

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## SUMMARY

AI Engineer with 2.5 years of experience building scalable AI applications using Python, FastAPI, OpenAI, Gemini, and vector databases. Experienced in developing RAG pipelines, conversational AI systems, voice assistants, and document intelligence platforms. Skilled in LLM integration, semantic search, WebSocket-based real-time systems, and AI backend architecture.

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## WORK EXPERIENCE

### Anviam Solutions Pvt. Ltd.

#### AI/ML Engineer

January, 2024 – Present

- Designed and implemented **RAG-based document intelligence pipelines** using **FAISS**, vector embeddings, and semantic search for high-accuracy retrieval across 500+ enterprise tenders, contracts, and proposal documents.
  - Built scalable document ingestion and processing pipelines supporting PDF parsing, chunking, metadata extraction, embedding generation, and semantic indexing workflows.
  - Integrated **Google Gemini (Vertex AI)** and **OpenAI LLMs** to generate structured, context-aware proposal responses, reducing manual effort in bid preparation and compliance analysis.
  - Engineered scalable FastAPI backend services for real-time AI processing, **asynchronous workflows**, and multi-user conversational systems.
  - Developed **Talking Bird**, a multi-agent Conversational AI platform integrating **SQL Agents** and **RAG Agents** for intelligent querying across structured and unstructured enterprise data.
  - Implemented semantic retrieval architectures using vector embeddings and **Weaviate** to improve natural language search accuracy across hybrid datasets.
  - Developed a real-time voice-based Conversational AI system using **ElevenLabs APIs** for automated user assessment and engagement workflows.
  - Applied **prompt engineering**, context **orchestration**, and conversational memory management techniques to improve response quality, dialogue consistency, and user interaction experience in AI-driven systems.
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## PROJECTS

### TenderQ

- Developed an AI-powered Tender & **RFP automation** platform using FastAPI, Google Gemini (Vertex AI), and RAG architecture for automated requirement extraction and **context-aware** response generation from enterprise documents.
- Engineered scalable document processing pipelines for PDF, DOCX, and XLSX files using Python, implementing text extraction, chunking, vector embeddings, and semantic retrieval with **FAISS**, **MongoDB**, and **AWS S3**.
- Built intelligent requirement analysis and validation workflows using Google Gemini LLMs for natural language understanding, **multi-language processing**, and automated proposal generation aligned with RFP compliance requirements.

### UIMS

- Developed a FastAPI-based **AI mental health platform** with secure JWT authentication, **multi-session chat management**, and role-based access for students, faculty, and administrators.
- Integrated **OpenAI LLMs** to enable conversational AI features, automated session summarization, and risk classification workflows for prioritizing critical student interactions.
- Built **natural-language-to-database pipelines** using **MongoDB** and **PostgreSQL** with LLM-based query routing, read-only query execution, and analytics APIs for session monitoring and reporting.

## Talking Bird

- Developed a **Conversational AI** platform using **ElevenLabs**, integrating SQL Agents and RAG Agents for intelligent querying across structured and unstructured data sources.
  - Implemented semantic search pipelines using **Weaviate** and vector embeddings for high-accuracy similarity retrieval and contextual conversation understanding.
  - Designed **LLM-driven conversational workflows** for intelligent query routing, context management, and real-time contextual response generation.
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## TECHNICAL SKILLS

- **AI/LLM Engineering:** LLMs, RAG, Prompt Engineering, AI Agents, Multi-Agent Systems, Conversational AI, Semantic Search, Embedding Pipelines
  - **Frameworks & Backend:** FastAPI, Flask, LangChain, LangGraph, REST APIs, WebSockets, Async Programming, Microservices Architecture
  - **AI APIs & Vector Databases:** OpenAI API, Gemini API, Vertex AI, Anthropic Claude API, Apollo API, InstantlyAI API, ElevenLabs API, FAISS, ChromaDB, Weaviate
  - **Machine Learning & Tools:** PyTorch, TensorFlow, Hugging Face, OpenCV, Scikit-learn, Docker, Git, Ubuntu, Postman, Swagger, n8n
  - **Databases & Cloud:** MySQL, MongoDB, PostgreSQL, AWS S3
  - **Core Domains:** Generative AI, NLP, Document Intelligence, Voice AI, Conversational AI, Computer Vision, Real-time AI Systems
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## EDUCATION

**Bachelor of Technology (B.Tech) – Computer Science Engineering (IoT & Cyber Security) | 2020 – 2024**

**Gulzar Group of Institutions | Affiliated to IK Gujral Punjab Technical University | Ludhiana**

CGPA: **8.4/10**

**Higher Secondary Education (12th) | 2019 – 2020**

**Government Senior Secondary School | Ludhiana**

Percentage: **85.6%**

**Matriculation (10th) | 2017 – 2018**

**SGD Public School | Ludhiana**

Percentage: **94.15%**

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